

Task

Implement CRUD operations using Python and Pandas on any dataset.

Data loading

Subtask:

Load a dataset into a pandas DataFrame.

Import pandas and load a dataset into a DataFrame, then display the first few rows.

```
import pandas as pd

# Using a publicly available dataset (Titanic)
url =
"https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv"
df = pd.read_csv(url)

df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

Create

Subtask:

Implement a function to add new records to the DataFrame.

Define a function to add a new record to the DataFrame using `pd.concat`.

```
def add_record(df, new_record_data):
    """Adds a new record to the DataFrame.
```

Args:

df: The original pandas DataFrame.

new_record_data: A dictionary representing the new record.

Returns:

The updated DataFrame with the new record added.

"""

```
new_record_df = pd.DataFrame([new_record_data])
```

```
updated_df = pd.concat([df, new_record_df], ignore_index=True)
```

```
return updated_df
```

Example usage:

```
new_data = {'PassengerId': 892, 'Survived': 1, 'Pclass': 3, 'Name':  
'Test, Mr. New Record',  
            'Sex': 'male', 'Age': 30, 'SibSp': 0, 'Parch': 0, 'Ticket':  
'12345', 'Fare': 10.0,  
            'Cabin': None, 'Embarked': 'S'}
```

```
df_updated = add_record(df, new_data)
```

```
display(df_updated.tail())
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.00	B42	S
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.45	NaN	S
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.00	C148	C
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.75	NaN	Q
891	892	1	3	Test, Mr. New Record	male	30.0	0	0	12345	10.00	None	S

Read

Subtask:

Implement a function to read and display records from the DataFrame.

Define the `read_records` function as instructed and then call it with `df_updated`.

```
def read_records(df):
```

```
    """Reads and displays records from the DataFrame.
```

Args:

df: The input pandas DataFrame.

```
    """
```

```
display(df.head())
```

```
read_records(df_updated)
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

Update

Subtask:

Implement a function to update existing records in the DataFrame.

Define the `update_record` function as requested and provide an example usage to demonstrate its functionality.

```
def update_record(df, index_to_update, updated_data):
    """Updates an existing record in the DataFrame.

    Args:
        df: The original pandas DataFrame.
        index_to_update: The index of the record to update.
        updated_data: A dictionary containing the updated values for the
            record.

    Returns:
        The updated DataFrame.
    """
    df.loc[index_to_update] = updated_data
    return df

# Example usage:
# Update the new record that was added in the previous step (index 891)
index_to_update = 891
updated_data = {'PassengerId': 892, 'Survived': 0, 'Pclass': 2, 'Name':
    'Updated, Mr. Test Record',
                'Sex': 'female', 'Age': 35, 'SibSp': 1, 'Parch': 1,
    'Ticket': '54321', 'Fare': 20.0,
                'Cabin': 'D10', 'Embarked': 'C'}

df_updated = update_record(df_updated, index_to_update, updated_data)
```

```
display(df_updated.tail())
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.00	B42	S
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.45	NaN	S
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.00	C148	C
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.75	NaN	Q
891	892	0	2	Updated, Mr. Test Record	female	35.0	1	1	54321	20.00	D10	C

Delete

Subtask:

Implement a function to delete records from the DataFrame.

Define the `delete_record` function, delete the record at index 891, and display the tail of the DataFrame to verify the deletion.

```
def delete_record(df, index_to_delete):  
    """Deletes a record from the DataFrame.
```

Args:

df: The original pandas DataFrame.

index_to_delete: The index of the record to delete.

Returns:

The updated DataFrame with the record deleted.

```
"""
```

```
df.drop(index_to_delete, axis=0, inplace=True)
```

```
return df
```

```
# Demonstrate the usage of the delete_record function by deleting the  
record at index 891
```

```
df_updated = delete_record(df_updated, 891)
```

```
# Display the tail of the DataFrame to verify that the record has been  
deleted
```

```
display(df_updated.tail())
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.00	NaN	S
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.00	B42	S
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.45	NaN	S
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.00	C148	C
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.75	NaN	Q

Summary:

- A new record was successfully added to the DataFrame using `pd.concat()`, increasing the number of rows.
- Existing records can be effectively read and displayed using standard pandas methods like `df.head()`.
- An existing record was successfully updated at a specific index (891) using the `.loc` accessor.
- A record at a specific index (891) was successfully deleted from the DataFrame using the `.drop()` method with `inplace=True`.
- The implemented functions provide a basic framework for CRUD operations on a pandas DataFrame, which can be extended for more complex data manipulation tasks.